XOSS Live Dashboard – User Manual

System Documentation

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1 Dashboard Parameter Configuration

Before launching the system, you can customize the operational parameters in the `config.json` file if needed[cite: 1]. Available settings include:

- **Save Interval:** Time interval (in seconds) between database recordings[cite: 1].
- **Dashboard Refresh Rate:** Frequency of live interface updates[cite: 1].
- **Maximum Sensors:** Maximum number of simultaneous Bluetooth connections allowed[cite: 1].
- **Auto-Export Interval:** Time interval (in seconds) for automatic JSON data exports[cite: 1].



Figure 1: Configuration parameters in `config.json`[cite: 1].

2 Launching the Application

Upon starting the application, the main Live Dashboard will automatically open in your web browser. The top control bar features three main action buttons:

1. **Scan Sensors:** Searches for nearby active Bluetooth heart rate straps[cite: 1, 2].
2. **Save JSON per Sensor:** Exports the recorded session data directly into the program's local `exports` folder[cite: 1, 2].
3. **Download JSON:** Downloads a consolidated JSON file directly to your computer's default *Downloads* directory[cite: 1, 2].

Note: Automatic File Generation

When the application initializes and begins recording data, it automatically generates the `xoss_readings.sqlite3` database, the `users.json` profile configuration file, and the `exports/` directory alongside the executable[cite: 1].

If the application has been used in previous sessions, historical hourly averages stored in the database will be displayed in the summary table at the bottom of the dashboard, as shown below[cite: 1, 2]:

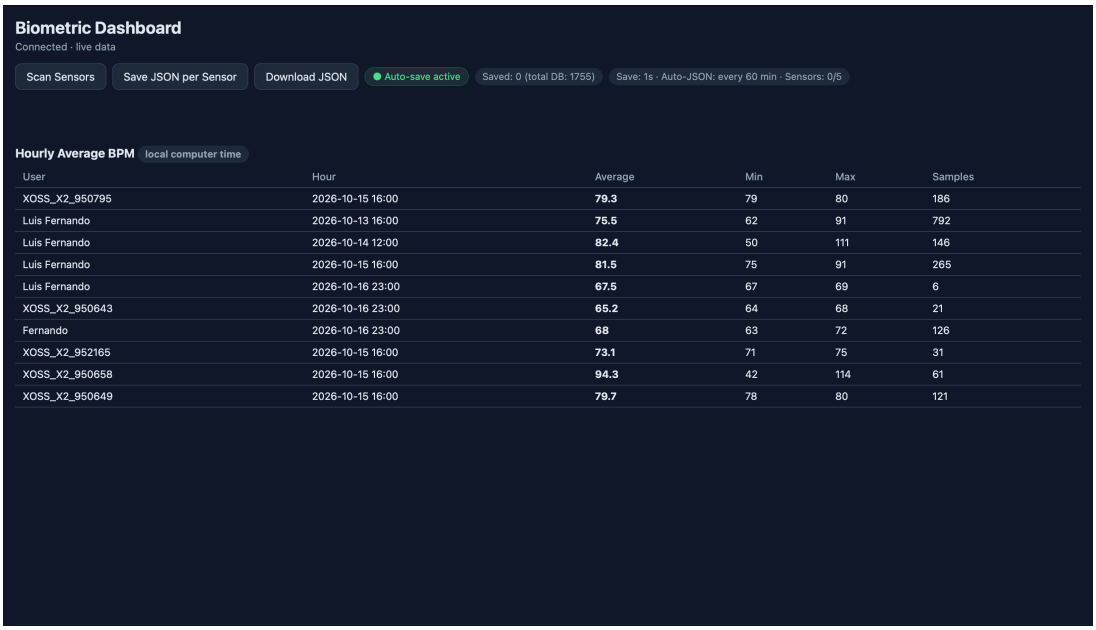


Figure 2: Main dashboard view and historical hourly BPM table[cite: 2].

3 Sensor Pairing and Activation

3.1 Sensor Placement and Activation

XOSS heart rate sensors are skin-activated. Before clicking **Scan Sensors**, ensure the strap is properly worn and making direct contact with the user's skin. If the strap is not worn or is positioned incorrectly, the sensor will remain powered off and will not appear in the scan results[cite: 2].

3.2 Selecting a Sensor

Once the scan completes, detected devices will be listed as buttons showing the prefix `XOSS_X2_` followed by a unique hardware ID (MAC address). Click on the desired sensor button to establish a connection and immediately begin data collection[cite: 2].

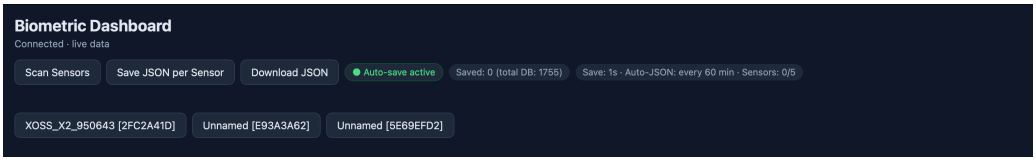


Figure 3: Available XOSS sensors detected during scanning.

4 Biometric Monitoring and User Profile

As soon as a sensor connects, real-time biometric tracking and automatic database logging begin. For accurate calorie expenditure calculations and clear file identification, update the user profile fields directly on the sensor's card:

- **User Name**[cite: 1, 2]
- **Weight (kg)**
- **Age**
- **Gender (Male / Female)**

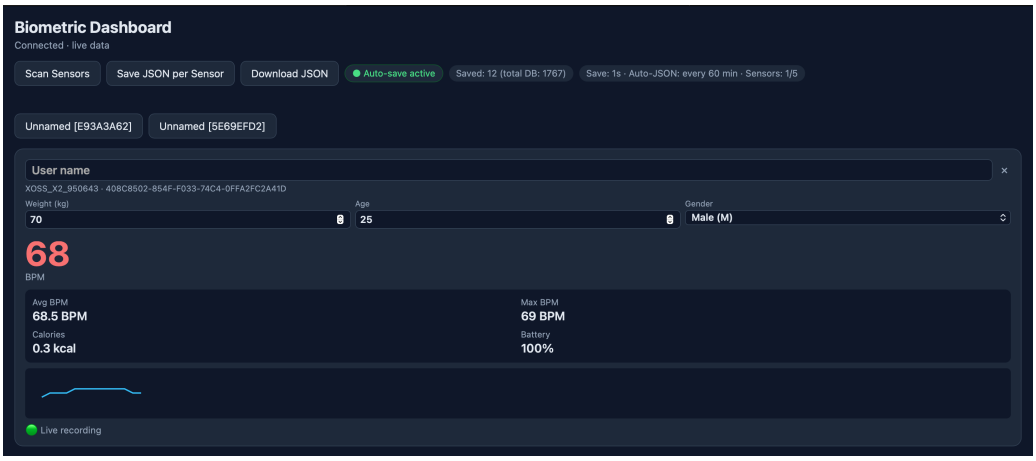
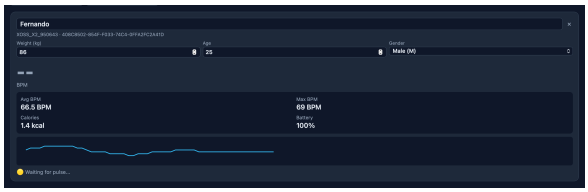


Figure 4: Live biometric monitoring and user profile configuration.

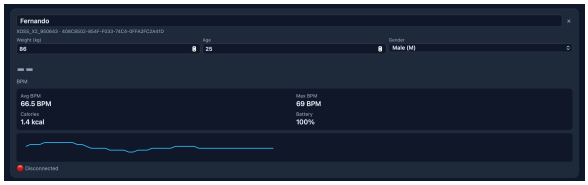
4.1 Connection Status Indicators

Each sensor card displays a real-time status indicator at the bottom left:

- **Green (`Live recording`):** The sensor is connected, actively reading heart rate data, and logging to the database.
- **Yellow (`Waiting for pulse...`):** The sensor is connected, but no skin contact is detected or the strap has entered standby mode[cite: 1, 2].
- **Red (`Disconnected`):** Bluetooth connection to the sensor was lost[cite: 2].



(a) Standby / Waiting for pulse state.



(b) Active / Disconnected states.

Figure 5: Sensor status indicators.

5 Data Storage and Troubleshooting

5.1 JSON Export Format

All exported data is saved in JSON format inside the `exports/` folder[cite: 1]. Files are automatically named per sensor using the following convention:

`[UserName]_[Hour-Minute]_[ExportNumber].json`
(Example: `Alex_14-30_1.json`)

5.2 Troubleshooting Disconnections

Sensor Reconnection Issue

If a sensor unexpectedly disconnects and cannot be re-added via **Scan Sensors**:

1. Click **Save JSON per Sensor** (or **Download JSON**) as a precautionary measure to back up your current session data.
2. Close and restart the application.
3. Repeat the scanning and connection steps[cite: 2].

Note: Recorded readings are continuously committed to the local SQLite database (`xoss_readings.sqlite3`) and will not be lost upon restarting, but manual export is recommended prior to closing[cite: 1].